

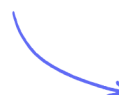
Multiple Choice Questions

Amount of Substance

Concentration Calculations / Reacting Mass Calculations / Reacting Volume Calculations / Calculations of Product

Easy (1 question)	/1
Medium (18 questions)	/22
Hard (2 questions)	/2
Total Marks	/25

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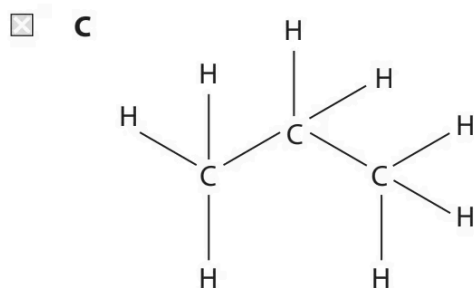
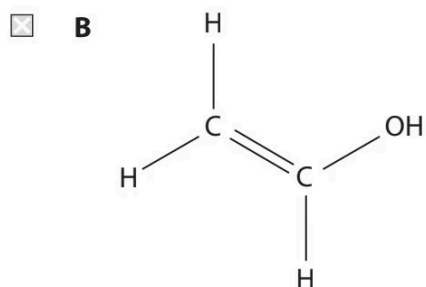
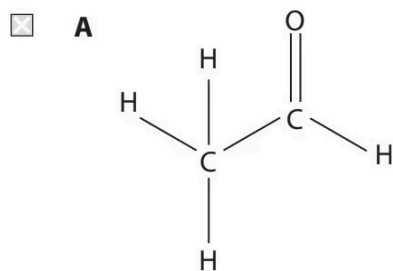


Easy Questions

- 1 The high resolution mass spectrum of a compound X has a molecular ion peak at $m/z = 44.0632$. Accurate relative atomic masses are given in the table.

Element	Relative atomic mass
Hydrogen	1.0079
Carbon	12.0000
Oxygen	15.9949

Which of these compounds, with a relative molecular mass of 44, gives rise to this peak?



(1 mark)

Medium Questions

- 1 (a) The maximum permitted concentration of sulfur in diesel fuel is 10 mg of sulfur in 1 kg of diesel fuel.

What is this concentration of sulfur in ppm?

- A. 0.00001
- B. 0.01
- C. 10
- D. 10000

(1 mark)

- (b) 3.2 kg of this diesel fuel is burned in air.

What is the maximum volume, in dm^3 , of sulfur dioxide which can be produced, measured at room temperature and pressure (r.t.p.)?

[Molar volume of a gas at r.t.p. = $24 \text{ dm}^3 \text{ mol}^{-1}$]

- A. 0.024
- B. 0.77
- C. 2.4
- D. 24

(1 mark)

- 2** A mass of 4.179 g of hydrated iron(III) sulfate, $\text{Fe}_2(\text{SO}_4)_3 \cdot \text{H}_2\text{O}$, was dissolved in deionised water and the solution made up to 200 cm^3 .

What is the concentration of sulfate ions, SO_4^{2-} , in the solution, in mol dm^{-3} ?

[Molar mass of $\text{Fe}_2(\text{SO}_4)_3 \cdot \text{H}_2\text{O} = 417.9 \text{ g mol}^{-1}$]

A. 0.01

B. 0.05

C. 0.10

D. 0.15

(1 mark)

- 3 (a)** A mass of 0.23 g of sodium was added to 350 cm³ water to form hydrogen and a solution of sodium hydroxide.



What is the concentration, in mol dm⁻³, of sodium hydroxide in the solution formed?

- A.** 0.010
- B.** 0.029
- C.** 0.29
- D.** 0.66

(1 mark)

- (b)** What is the maximum volume, in cm³, of hydrogen which could be formed, measured at r.t.p.?

[Molar volume of a gas at r.t.p. = 24 dm³ mol⁻¹]

- A.** 120
- B.** 240
- C.** 480
- D.** 2800

(1 mark)

- (c)** The sodium hydroxide solution was neutralised with sulfuric acid.

Which is the ionic equation for this reaction?

- A.** $\text{H}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{H}_2\text{O (l)}$
- B.** $\text{SO}_4^{2-}(\text{aq}) + 2\text{Na}^+(\text{aq}) \rightarrow \text{Na}_2\text{SO}_4(\text{aq})$
- C.** $\text{H}_2\text{SO}_4(\text{aq}) + 2\text{Na}^+(\text{aq}) + 2\text{OH}^-(\text{aq}) \rightarrow \text{Na}_2\text{SO}_4(\text{aq}) + 2\text{H}_2\text{O (l)}$
- D.** $2\text{H}^+(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) + 2\text{Na}^+(\text{aq}) + 2\text{OH}^-(\text{aq}) \rightarrow 2\text{Na}^+(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) + 2\text{H}_2\text{O (l)}$

(1 mark)

- (d) Sodium hydroxide solution was added to magnesium sulfate solution. The equation for the reaction is shown.



What is the atom economy (by mass) for the production of magnesium hydroxide?

[A_r values: H = 1.0 O = 16.0 Na = 23.0 Mg = 24.3 S = 32.1]

- A. 29.1%
- B. 41.0%
- C. 48.4%
- D. 50.0%

(1 mark)

- 4 Which conversion has the **lowest** percentage atom economy (by mass) for the formation of CaCl_2 ?

- A. $\text{Ca} + \text{Cl}_2 \rightarrow \text{CaCl}_2$
- B. $\text{Ca} + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2$
- C. $\text{CaCO}_3 + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2\text{O} + \text{CO}_2$
- D. $\text{CaCO}_3 + 2\text{NaCl} \rightarrow \text{CaCl}_2 + \text{Na}_2\text{CO}_3$

(1 mark)

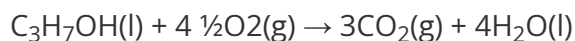
- 5 The concentration of nitrogen dioxide in a sample of air is 0.5 ppm.

What is the percentage of nitrogen dioxide molecules in this sample of air?

- A. 0.5%
- B. 0.005%
- C. 0.00005%
- D. 0.0000005%

(1 mark)

- 6 The equation for the complete combustion of propan-1-ol is shown.



2.00×10^{-3} mol of propan-1-ol undergoes complete combustion.

What mass of carbon dioxide is formed?

- A. 0.0293g
- B. 0.0880g
- C. 0.132g
- D. 0.264g

(1 mark)

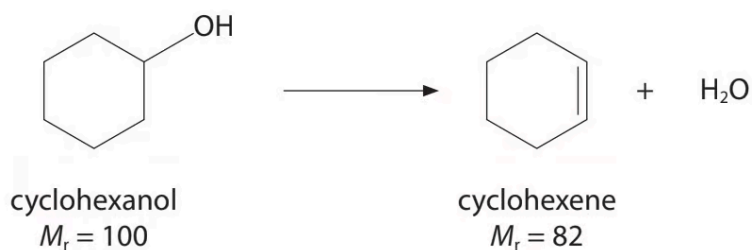
- 7 A student is provided with 25.0 cm^3 of 1.00 mol dm^{-3} hydrochloric acid.

What volume of distilled water should the student add to this solution to make a $0.0500 \text{ mol dm}^{-3}$ solution?

- A. 25.0 cm^3
- B. 50.0 cm^3
- C. 475 cm^3
- D. 500 cm^3

(1 mark)

- 8 Cyclohexene may be prepared by the dehydration of cyclohexanol.



What mass of cyclohexene can be made from 12.5 g of cyclohexanol if the yield is 51.2%?

- A. 5.25 g

- B.** 6.40 g
- C.** 7.80 g
- D.** 10.25 g

(1 mark)

9 Iron(III) oxide is reduced by hydrogen in a two-step process.

Step 1:	$3\text{Fe}_2\text{O}_3 + \text{H}_2 \rightarrow 2\text{Fe}_3\text{O}_4 + \text{H}_2\text{O}$
Step 2:	$\text{Fe}_3\text{O}_4 + 4\text{H}_2 \rightarrow 3\text{Fe} + 4\text{H}_2\text{O}$

What is the maximum mass of iron that could be produced from 39.9 tonnes of Fe_2O_3 ?

[A_r values: H = 1.0 O = 16.0 Fe = 55.8]

- A.** 6.98 tonnes
- B.** 13.95 tonnes
- C.** 27.90 tonnes
- D.** 41.85 tonnes

(1 mark)

10 Which of these gases occupies 6.0 dm^3 at room temperature and pressure (r.t.p.)?

[molar volume of gas at r.t.p. = $24.0 \text{ dm}^3 \text{ mol}^{-1}$]

A_r values: He = 4.0 C = 12.0 N = 14.0 O = 16.0]

- A.** 2.0 g of helium
- B.** 4.0 g of oxygen
- C.** 11.0 g of carbon dioxide
- D.** 14.0 g of nitrogen

(1 mark)

11 How many moles are there in 15.1 cm^3 of liquid propan-1-ol?

[Density of propan-1-ol = 0.80 g cm^{-3} M_r of propan-1-ol = 60]

- A.** $(0.80 \times 15.1) \div 60$
- B.** $0.80 \div (60 \times 15.1)$
- C.** $60 \div (0.80 \times 15.1)$
- D.** $(60 \times 15.1) \div 0.80$

(1 mark)

12 Which of these solutions contains the greatest number of ions?

- A.** 20.0 cm^3 of 0.5 mol dm^{-3} KCl
- B.** 0.40 dm^3 of 0.03 mol dm^{-3} KCl
- C.** 10.0 cm^3 of 0.6 mol dm^{-3} CaCl_2
- D.** 0.15 dm^3 of 0.04 mol dm^{-3} CaCl_2

(1 mark)

13 What volume, in dm^3 , of hydrogen gas will be produced when 3.00 g of lithium is reacted with water at room temperature and pressure (r.t.p.)?

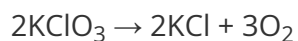


[Molar volume of gas at r.t.p. = $24.0 \text{ dm}^3 \text{ mol}^{-1}$]

- A.** 0.217
- B.** 0.435
- C.** 5.22
- D.** 10.4

(1 mark)

- 14** Potassium chlorate(V) decomposes on heating to form oxygen.



What is the atom economy (by mass) for the formation of oxygen?

[A_r values: O = 16.0 Cl = 35.5 K = 39.1]

- A.** 13.1%
- B.** 26.1%
- C.** 39.2%
- D.** 64.3%

(1 mark)

- 15** 10.00 g of hydrated magnesium sulfate, $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$, is heated to remove the water of crystallisation.

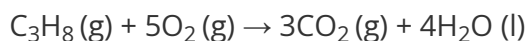
What mass of anhydrous magnesium sulfate, MgSO_4 , is formed?

[Molar mass of $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ = 246.4 g mol^{-1}]

- A.** 2.84 g
- B.** 4.89 g
- C.** 5.11 g
- D.** 7.16 g

(1 mark)

16 Propane burns completely in oxygen as shown.



100 cm³ of propane was mixed with 600 cm³ of oxygen and the mixture was ignited.

What is the **total** volume, in cm³, of the gas mixture at the end of the reaction?

All gas volumes were measured at room temperature and pressure.

A. 300

B. 400

C. 700

D. 800

(1 mark)

17 Barium chloride solution, BaCl₂(aq), reacts with gallium sulfate solution, Ga₂(SO₄)₃(aq) to form a precipitate of barium sulfate, BaSO₄(s).

What is the minimum volume of 0.100 mol dm⁻³ barium chloride needed to precipitate **all** the sulfate ions in 200 cm³ of 0.05 mol dm⁻³ gallium sulfate?

A. 100 cm³

B. 200 cm³

C. 300 cm³

D. 400 cm³

(1 mark)

18 Which aqueous solution contains the greatest number of **ions**?

A. 200 cm³ of 1.5 mol dm⁻³ MgCl₂

B. 400 cm³ of 0.8 mol dm⁻³ MgSO₄

C. 500 cm³ of 1.0 mol dm⁻³ NaCl

D. 1000 cm³ of 0.25 mol dm⁻³ Na₂SO₄

(1 mark)

Hard Questions

1 Which aqueous solution has the **highest** concentration, in mol dm^{-3} , of chloride ions?

- A. $0.1 \text{ g dm}^{-3} \text{ HCl}$
- B. $0.1 \text{ g dm}^{-3} \text{ NaCl}$
- C. $0.1 \text{ g dm}^{-3} \text{ KCl}$
- D. $0.1 \text{ g dm}^{-3} \text{ BaCl}_2$

(1 mark)

2 A 4.00 mol dm^{-3} solution of an acid is used to prepare dilute solutions of the acid.

What volume of water is required to make up 150 cm^3 of 0.35 mol dm^{-3} solution of the acid?

- A. 13.1 cm^3
- B. 52.5 cm^3
- C. 97.5 cm^3
- D. 136.9 cm^3

(1 mark)